

35.4 Goodness of fit

In the normal linear model the residual sum of squares both estimates σ^2 and measures fit, and its known distribution makes a fit too poor to be chance announce itself (Chapter 14). Binary data have no free dispersion parameter, so the deviance has only the second role, and often cannot fill it either.

35.4.1 Two statistics for grouped data

Definition 35.4.1. Pearson and deviance statistics

For grouped binomial data $y_i \sim \text{Binomial}(m_i, \pi_i)$, $i = 1, \dots, N$, with fitted probabilities $\hat{\pi}_i$ from a model with p parameters, the **Pearson statistic** and the **deviance** are

$$\begin{aligned} X^2 &= \sum_{i=1}^N \frac{(y_i - m_i \hat{\pi}_i)^2}{m_i \hat{\pi}_i (1 - \hat{\pi}_i)}, \\ D &= 2 \sum_{i=1}^N \left\{ y_i \log \frac{y_i}{m_i \hat{\pi}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - m_i \hat{\pi}_i} \right\}, \end{aligned} \quad (35.4.1)$$

with the convention $0 \log 0 = 0$. Both compare the fit with the **saturated** model that gives each group its own probability.

X^2 is the sum over all $2N$ cells of successes and failures of $(\text{observed} - \text{fitted})^2 / \text{fitted}$, and D is twice the sum of $\text{observed} \times \log(\text{observed}/\text{fitted})$: the classical categorical-data statistics, specialized to the binomial.

Proposition 35.4.1. What the two statistics measure

- (a) (*Grouped data.*) Fix N and the model, let $m_i \rightarrow \infty$ for every i with π_i bounded away from 0 and 1, and suppose the model holds. Then X^2 and D both converge in distribution to $\chi^2(N - p)$, and $X^2 - D \rightarrow 0$ in probability.
- (b) (*Ungrouped data, canonical link.*) If $m_i = 1$ for every i and $\hat{\beta}$ solves the logistic likelihood equations Eq. 35.1.2, then

$$D = -2 \sum_{i=1}^n \{ \hat{\pi}_i \log \hat{\pi}_i + (1 - \hat{\pi}_i) \log(1 - \hat{\pi}_i) \}. \quad (35.4.2)$$

The deviance is therefore a function of the fitted probabilities alone: it does not depend on which observations were successes, it is bounded above by $2n \log 2$, and it has no $\chi^2(n - p)$ limit.

- (c) (*Differences.*) For nested models $\omega \subset \Omega$ with q parameters between them, $D_\omega - D_\Omega$ does not depend on the grouping and converges to $\chi^2(q)$ under the smaller model, grouped or not.

Proof.

- (a) The statement about D is Theorem 34.5.5 specialized to the binomial family: with N fixed and every $m_i \rightarrow \infty$, the model is a smooth submodel of an N -parameter family

whose maximum likelihood estimates are asymptotically normal, so the likelihood ratio statistic against the saturated model has $N - p$ degrees of freedom. For X^2 , put $b = m_i \hat{\pi}_i$ and $r_i = (y_i - b) / \{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}^{1/2}$, so that $r_i = O_p(1)$ and $y_i - b = O_p(m_i^{1/2})$. The i th deviance summand is $g(y_i)$ for $g(a) = 2\{a \log(a/b) + (m_i - a) \log((m_i - a)/(m_i - b))\}$, and $g(b) = g'(b) = 0$ and $g''(b) = 2/\{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}$, while the third derivative is $O(m_i^{-2})$ near that point. Taylor's theorem therefore gives $g(y_i) = r_i^2 + O_p(m_i^{-1/2})$, using $\hat{\pi}_i$ bounded away from 0 and 1. Summing over the N fixed groups, $D - X^2 \rightarrow 0$ in probability, so X^2 has the same limit as D .

- (b) Write $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}$, so that $\hat{\eta}_i = \log \hat{\pi}_i - \log(1 - \hat{\pi}_i)$. For ungrouped data the saturated model sets $\hat{\pi}_i = y_i$ and has log-likelihood zero, so

$$\begin{aligned} D &= -2 \sum_i \{y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i)\} \\ &= -2 \sum_i \{y_i \hat{\eta}_i + \log(1 - \hat{\pi}_i)\} \\ &= -2 \left\{ \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} + \sum_i \log(1 - \hat{\pi}_i) \right\}. \end{aligned}$$

The likelihood equations say $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \hat{\boldsymbol{\pi}}$, so $\mathbf{X}^\top \mathbf{y}$ may be replaced by $\mathbf{X}^\top \hat{\boldsymbol{\pi}}$, giving $D = -2 \sum_i \{\hat{\pi}_i \hat{\eta}_i + \log(1 - \hat{\pi}_i)\}$, which is Eq. 35.4.2 after substituting for $\hat{\eta}_i$. Each summand of Eq. 35.4.2 is at most $2 \log 2$, attained at $\hat{\pi}_i = 1/2$, which gives the bound. For the last claim write $H(\pi) = -\{\pi \log \pi + (1 - \pi) \log(1 - \pi)\}$ and $f_n(\boldsymbol{\beta}) = 2 \sum_i H(\pi_i)$ with $\pi_i = \pi_i(\boldsymbol{\beta})$, so that $D = f_n(\hat{\boldsymbol{\beta}})$ identically: D is a smooth deterministic function of $\hat{\boldsymbol{\beta}}$, with no other dependence on the data. Now $dH/d\eta = -\eta \pi(1 - \pi)$, and $\pi(1 - \pi) \leq e^{-|\eta|}$, so $|dH/d\eta| \leq |\eta| e^{-|\eta|} \leq e^{-1}$ and $\|\nabla f_n\| \leq 2e^{-1} \sum_i \|\mathbf{x}_{(i)}\| = O(n)$ for bounded covariates. With $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = O_p(n^{-1/2})$ from Theorem 34.5.1, the mean value theorem gives $D - f_n(\boldsymbol{\beta}) = O_p(n^{1/2})$, so

$$n^{-1}D - 2n^{-1} \sum_{i=1}^n H(\pi_i) \rightarrow 0 \text{ in probability,}$$

whereas $n^{-1}\chi^2(n - p) \rightarrow 1$. The two agree only by accident: the simulation of Example (Three deviances that mean nothing) has $n^{-1}D$ averaging $206.8/200 = 1.03$ under one correct model and $263.6/200 = 1.32$ under another. So D has no $\chi^2(n - p)$ limit. The same bound leaves D fluctuating about $f_n(\boldsymbol{\beta})$ by $O_p(n^{1/2})$, which is the order of the standard deviations in Example (Three deviances that mean nothing); what fails is not the spread of D but its location.

- (c) The saturated log-likelihood is common to both deviances and cancels, so the difference is twice a log-likelihood ratio between two nested smooth models with q free parameters between them, and Theorem 34.5.5 applies with q fixed.

■

Example. Is the party scale linear on the logit scale?

Collapse the survey of Example (Voting intention in the 1996 American election) onto its seven party groups, of between 37 and 200 respondents each, and fit logit $\pi = \beta_0 + \beta_1$ party. The deviance is 12.765 on 5 degrees of freedom ($p = 0.0257$) and the Pearson statistic is 11.899 ($p = 0.0362$). The Pearson residuals locate the misfit: -2.067 at “independent-Democrat”, where the model predicts 0.132 against an observed 0.065, and 2.122 at “independent-

Republican”, where it predicts 0.640 against 0.745. The transition across the middle of the scale is sharper than a straight line in the log odds allows, and a quadratic term does not help: the deviance falls only to 12.762, so the misfit is a step, not a curve.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import stats

anes = sm.datasets.anes96.load_pandas().data
cells = anes.groupby("PID")["vote"].agg(["sum", "count"]).reset_index()
counts = np.column_stack([cells["sum"], cells["count"] - cells["sum"]])
Xg = sm.add_constant(cells[["PID"]])

g = sm.GLM(counts, Xg, family=sm.families.Binomial()).fit()
print(cells.to_string(index=False))
print(f"deviance {g.deviance:.3f} on {g.df_resid} df, p = {stats.chi2.sf(g.deviance,
                                g.df_resid):.4f}")
print(f"Pearson {g.pearson_chi2:.3f} on {g.df_resid} df, "
      f"p = {stats.chi2.sf(g.pearson_chi2, g.df_resid):.4f}")
```

[Run this code online](#)

The conditions in part (a) are not decoration. If N grows with n instead, as it does whenever a continuous covariate is in the model, the approximation fails, worst of all in the extreme case $m_i \equiv 1$ of part (b).

35.4.2 The ungrouped deviance measures nothing

Eq. 35.4.2 deserves to be read twice. The left-hand side looks like a discrepancy between data and fit; the right-hand side contains no data at all. It is twice the total Bernoulli entropy of the fitted probabilities, a measure of how *undecided* the model is: probabilities near 1/2 give a deviance near $2n \log 2$ and extreme ones give a small deviance, whether or not the model is correct.

Example. Three deviances that mean nothing

Simulate $n = 200$ observations with one covariate and an intercept in three ways: a correct model with a strong slope, a correct model with a weak slope, and a model that leaves a genuine quadratic effect out of the fitted predictor. All three fit two parameters, so $n - p = 198$ and a chi-squared reference would have mean 198 and standard deviation 19.9. Over 2000 replications:

data-generating mechanism	mean deviance	standard deviation
correct, strong slope	206.8	14.3
correct, weak slope	263.6	7.0
quadratic omitted	252.2	8.5

Two *correct* models with the same n and p differ by nearly sixty in mean deviance, and the badly misspecified model has a *smaller* deviance than one of them: no reference distribution can serve all three (Figure 35.4.1).

The survey model of Example (Voting intention in the 1996 American election) has ungrouped deviance 528.29 on 939 degrees of freedom; dropping every covariate but education gives an obviously wrong model with deviance 1275.16 on 942. Neither number can be referred to a table.

```
# the ungrouped deviance is a function of the fitted probabilities alone
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"].to_numpy()
fit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
p = fit.fittedvalues.to_numpy()

entropy = -2 * np.sum(p * np.log(p) + (1 - p) * np.log(1 - p))
print("deviance reported ", round(fit.deviance, 6))
print("entropy of the fit ", round(entropy, 6))
```

[Run this code online](#)

```
def deviance_null(n, B, seed=0):
    """Ungrouped deviances under three data-generating mechanisms: a correct model
    with a
    strong slope, a correct model with a weak slope, and a model that omits a real
    quadratic
    effect. All three fit an intercept and one slope to n observations."""
    rng = np.random.default_rng(seed)
    out = np.empty((B, 3))
    for b in range(B):
        x = rng.normal(size=n)
        Z = np.column_stack([np.ones(n), x])
        for j, eta in enumerate([0.4 + 1.5 * x, 0.4 + 0.3 * x, 0.4 + 1.5 * x - 1.8 *
                                x ** 2]):
            yb = (rng.random(n) < 1 / (1 + np.exp(-eta))).astype(float)
            out[b, j] = sm.GLM(yb, Z, family=sm.families.Binomial()).fit().deviance
    return out

demo = deviance_null(200, 100, seed=1)
print("n = 200 and p = 2, so a chi-squared reference would have mean 198 and "
      f"standard deviation {np.sqrt(2 * 198):.1f}")
for j, label in enumerate(["correct, strong slope", "correct, weak slope ",
                           "quadratic omitted "]):
    print(f"{label}: mean {demo[:, j].mean():6.1f}, "
          f"standard deviation {demo[:, j].std():4.1f}")
```

[Run this code online](#)

The Pearson statistic escapes Eq. 35.4.2, since $X^2 = \sum_i \{y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i)\} - n$ does depend on $\mathbf{y}$, but gains nothing by it (Exercise (B1)). What survives is part (c): for ungrouped data, test the model against enlargements of it rather than against the saturated alternative, and use AIC or cross-validation (Chapter 29) when the comparisons are not nested.

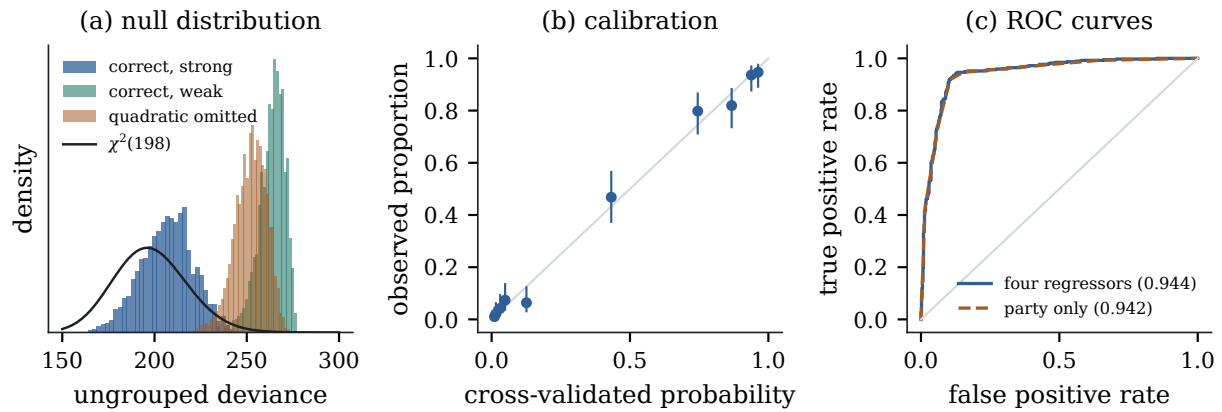


Figure 35.4.1: (a) Simulated null distributions of the ungrouped deviance with $n = 200$ and two fitted parameters, under the three mechanisms of [Example \(Three deviances that mean nothing\)](#), against the $\chi^2(198)$ density. (b) Cross-validated calibration of the survey model in ten bins, with pointwise intervals. (c) ROC curves for the four-regressor model and for party identification alone.

35.4.3 Grouping by fitted values

Hosmer and Lemeshow (1980) proposed making grouped data out of ungrouped: sort by $\hat{\pi}_i$, cut into G groups of roughly equal size, and compare

$$\hat{C} = \sum_{k=1}^G \frac{(O_k - E_k)^2}{E_k(1 - E_k/n_k)}, \quad O_k = \sum_{i \in k} y_i, \quad E_k = \sum_{i \in k} \hat{\pi}_i,$$

referred to $\chi^2(G - 2)$. The reference distribution has no proof behind it: it was established by simulation, and the two lost degrees of freedom are an empirical finding. The statistic is a useful descriptive check and an unstable test.

Example. The same model, three verdicts

For the four-regressor model of Example (Voting intention in the 1996 American election):

groups G	$\hat{C}$	p
8	8.670	0.193
10	12.232	0.141
12	16.119	0.096

The p -value halves as G goes from 8 to 12, and different software makes different choices about G , about how to break ties in $\hat{\pi}_i$, and about whether to use equal-sized groups or equally spaced cut-points. None of these choices is more principled than any other.

```
def hosmer_lemeshow(y, p, groups):
    """The Hosmer-Lemeshow statistic and its nominal p-value with the given number
    of groups."""
    order = np.argsort(p)
    chunks = np.array_split(order, groups)
    stat = 0.0
    for c in chunks:
```

```

    e = p[c].sum()
    stat += (y[c].sum() - e) ** 2 / (e * (1 - e / len(c)))
    return stat, stats.chi2.sf(stat, groups - 2)

for gr in (8, 10, 12):
    stat, pv = hosmer_lemeshow(y, p, gr)
    print(f"{gr:2d} groups: statistic {stat:6.3f}, p = {pv:.3f}")

```

[Run this code online](#)

35.4.4 Calibration and discrimination

Two questions hide inside “does the model fit”. **Calibration** asks whether the fitted probabilities are right: among observations with $\hat{\pi}_i \approx 0.3$, do about thirty per cent have $y_i = 1$? **Discrimination** asks whether they order the observations correctly, and a model can have either without the other. Classification tables of sensitivity against specificity answer a third question, and depend on a threshold and on a loss they rarely state; pseudo- R^2 measures such as $1 - D/D_{\text{null}}$ have no interpretation of their own, and by [Proposition 35.4.1\(b\)](#) the ungrouped version of that ratio is a ratio of entropies.

Proposition 35.4.2. In-sample calibration is automatic

Let $\hat{\beta}$ solve Eq. 35.1.2 with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, let $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$, and suppose $\hat{\eta}$ is not constant. Fit the two-parameter logistic model $\text{logit } \pi_i = a + b\hat{\eta}_i$ to the same responses. Then $\hat{a} = 0$ and $\hat{b} = 1$ exactly.

Proof. The likelihood equations of the two-parameter model at $(a, b) = (0, 1)$ are $\sum_i (y_i - \hat{\pi}_i) = 0$ and $\sum_i \hat{\eta}_i (y_i - \hat{\pi}_i) = 0$. The first holds because $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \mathbf{0}$; the second because $\hat{\boldsymbol{\eta}} = \mathbf{X}\hat{\boldsymbol{\beta}} \in \mathcal{C}(\mathbf{X})$, so $\hat{\boldsymbol{\eta}}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = 0$. So $(0, 1)$ solves the two-parameter likelihood equations; it remains to see that a maximizer exists. The design $[\mathbf{1}, \hat{\boldsymbol{\eta}}]$ has rank two, since $\hat{\eta}$ is not constant. Suppose some $(a, b) \neq (0, 1)$ had $s_i(a + b\hat{\eta}_i) \geq 0$ for every i . As $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ there is an $\mathbf{e}$ with $\mathbf{X}\mathbf{e} = \mathbf{1}$, and $\mathbf{b}^* = a\mathbf{e} + b\hat{\boldsymbol{\beta}}$ satisfies $\mathbf{X}\mathbf{b}^* = a\mathbf{1} + b\hat{\boldsymbol{\eta}} \neq \mathbf{0}$, so $\mathbf{b}^* \neq \mathbf{0}$ separates $\mathbf{X}$ — contradicting the overlap that the finiteness of $\hat{\boldsymbol{\beta}}$ guarantees (Theorem 35.3.1). The two-parameter data are therefore in overlap, and Theorem 35.3.1(c) gives a unique maximizer, which must be the stationary point $(0, 1)$. ■

The **calibration slope** $\hat{b}$ is therefore worthless on the data that produced $\hat{\boldsymbol{\beta}}$. It becomes informative when $\hat{\eta}$ comes from a held-out fold or a new sample, where $\hat{b} < 1$ is the signature of overfitting. On the survey model of Example (Voting intention in the 1996 American election), ten-fold cross-validated predictions give a slope of 0.983 and an intercept of 0.005: four regressors and 944 observations is not a regime in which overfitting bites.

Proposition 35.4.3. What the area under the ROC curve is, and is not

Let $\hat{s}_i$ be any scores, let I be a uniformly chosen index with $y_I = 1$ and J an independent uniformly chosen index with $y_J = 0$. The area under the curve traced by (false positive rate,

true positive rate) as a threshold moves through the scores equals

$$\text{AUC} = \Pr(\hat{s}_I > \hat{s}_J) + \frac{1}{2} \Pr(\hat{s}_I = \hat{s}_J),$$

the normalized Mann–Whitney statistic. Consequently the AUC depends on the scores only through their ranks, so it is unchanged by every strictly increasing transformation of $\hat{s}$, and in particular it says nothing whatever about calibration.

Proof. Order the distinct score values downwards. Moving the threshold past one of them advances the curve by the fraction ΔFPR of controls and the fraction ΔTPR of cases at that value, adding the trapezoid $\Delta\text{FPR}\{\text{TPR} + \frac{1}{2}\Delta\text{TPR}\}$. Summing, the first part counts pairs with $\hat{s}_I > \hat{s}_J$ and the second counts ties with weight $\frac{1}{2}$, each divided by n_1n_0 . Invariance is immediate, since an increasing transformation preserves every comparison $\hat{s}_I > \hat{s}_J$. ■

So a model whose probabilities are all wrong by a factor of ten has the same AUC as the correctly calibrated model with the same ordering. The AUC is insensitive in the other direction too: for Example (Voting intention in the 1996 American election) party identification alone gives $\text{AUC} = 0.942$, and adding age, education and income raises it to 0.944. An increment of 0.002 in a rank-based summary is the wrong scale on which to judge the three covariates.

35.4.5 Residuals and influence

The diagnostics of [Chapter 20](#) carry over, with the working weights as the metric.

Definition 35.4.2. Residuals for binomial data

With $\hat{w}_i = m_i\hat{\pi}_i(1-\hat{\pi}_i)$, $\hat{\mathbf{W}} = \text{diag}(\hat{w}_i)$ and $\hat{h}_i$ the i th diagonal entry of $\hat{\mathbf{W}}^{1/2}\mathbf{X}(\mathbf{X}^\top\hat{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}^\top\hat{\mathbf{W}}^{1/2}$, the **Pearson** and **deviance** residuals are

$$r_i = \frac{y_i - m_i\hat{\pi}_i}{\sqrt{\hat{w}_i}}, \quad d_i = \text{sign}(y_i - m_i\hat{\pi}_i)\sqrt{D_i},$$

where D_i is the i th summand of the deviance in [Eq. 35.4.1](#). Their **standardized** forms divide by $(1 - \hat{h}_i)^{1/2}$.

By construction $\sum_i r_i^2 = X^2$ and $\sum_i d_i^2 = D$. The matrix in the definition is the hat matrix of the last weighted least squares step of [Proposition 35.1.3](#), so it is symmetric and idempotent with trace $\text{rank}(\mathbf{X})$, and [Section 6.8](#) applies to it unchanged. The analogue of Cook's distance ([Definition 20.4.1](#)) is $r_i^2\hat{h}_i/\{p(1 - \hat{h}_i)^2\}$.

Two things change. For ungrouped data r_i takes only two values once $\hat{\pi}_i$ is given, so plots against a covariate show two curves rather than a cloud and must be smoothed or binned. And $\hat{h}_i$ carries the factor $\hat{w}_i$, so high leverage now means a covariate vector that is extreme *and* a fitted probability near 1/2; a point whose outcome the model regards as certain has low leverage wherever its covariates lie, the opposite of the linear-model intuition.

35.4.6 Exercises

35.4.6.1 A. Check your understanding

Exercise. A1

Use Eq. 35.4.2 to find the largest and smallest possible values of the ungrouped deviance of a logistic fit, and describe the fits that attain them.

Solution. Each summand $-2\{\hat{\pi} \log \hat{\pi} + (1 - \hat{\pi}) \log(1 - \hat{\pi})\}$ lies in $[0, 2 \log 2]$, so $0 \leq D \leq 2n \log 2$. The upper bound is attained when every $\hat{\pi}_i = 1/2$, as for the intercept-only model fitted to equal numbers of successes and failures. The lower bound is approached only as the fitted probabilities go to 0 and 1, which by Theorem 35.3.1 needs complete separation, so 0 is an infimum and not a minimum.

35.4.6.2 B. Practice

Exercise. B1

Show that for ungrouped data $X^2 = \sum_i \{y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i)\} - n$, and deduce that for the intercept-only model $X^2 = n$ exactly, whatever the data. What does that say about X^2 as a test of fit?

Solution. With $m_i = 1$, $(y_i - \hat{\pi}_i)^2 / \{\hat{\pi}_i(1 - \hat{\pi}_i)\}$ equals $(1 - \hat{\pi}_i)/\hat{\pi}_i$ when $y_i = 1$ and $\hat{\pi}_i/(1 - \hat{\pi}_i)$ when $y_i = 0$, and both equal $y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i) - 1$; summing gives the identity. For the intercept-only model $\hat{\pi}_i = \bar{y}$ for every i , so $X^2 = n\bar{y}/\bar{y} + n(1 - \bar{y})/(1 - \bar{y}) - n = n$. A statistic that equals the sample size for every data set cannot test anything.

Exercise. B2

Let $\hat{\eta}$ be fitted on one sample and evaluated on an independent sample from the same population. Show that the calibration slope b has probability limit 1 if the model is correct and $\hat{\beta}$ is consistent, and explain why a value below one indicates overfitting.

Exercise. B3

Suppose the covariate is $N(\mu_1, 1)$ among cases and $N(\mu_0, 1)$ among controls. Compute the AUC of the correctly specified logistic model in terms of $\Delta = \mu_1 - \mu_0$, and find the Δ needed for an AUC of 0.9.

Solution. The fitted score is monotone in the covariate, so the AUC is $\Pr(X_1 > X_0) = \Phi(\Delta/\sqrt{2})$, since $X_1 - X_0 \sim N(\Delta, 2)$. Setting this to 0.9 gives $\Delta = \sqrt{2} \Phi^{-1}(0.9) = 1.812$. An AUC of 0.65 needs only $\Delta \approx 0.55$, a difference highly significant in a large sample yet nearly useless for classifying an individual.

35.4.6.3 C. Going deeper

Exercise. C1

Example (The same model, three verdicts) shows the Hosmer–Lemeshow statistic moving with the number of groups. This exercise holds G fixed and moves only the tie-breaking. Fit the intercept-only logistic model to $n = 100$ ungrouped observations of which exactly 50 are successes, and take $G = 10$ groups of 10.

- Show that $\hat{\pi}_i = \frac{1}{2}$ for every i , and hence that $E_k = 5$, $n_k = 10$ and $E_k(1 - E_k/n_k) = \frac{5}{2}$ for every group, whatever the grouping, so that $\hat{C} = \frac{2}{5} \sum_{k=1}^{10} (O_k - 5)^2$.
- Exhibit a grouping giving $\hat{C} = 0$ and one giving $\hat{C} = 100$, and give the nominal p -values against $\chi^2(8)$ in words.
- Explain why both are legitimate outputs of “sort by $\hat{\pi}_i$ and cut into ten groups of equal size”, and what this says about $\hat{C}$ as a function of the data and the model.
- The model in (a) is correct if the successes really are independent with probability $\frac{1}{2}$. Does that matter to the argument?

Solution.

- With only an intercept, $\hat{\pi}_i = \bar{y} = \frac{1}{2}$ for every i (**Proposition 35.4.2** gives the same conclusion, since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ forces $\sum_i (y_i - \hat{\pi}_i) = 0$). Then $E_k = \sum_{i \in k} \hat{\pi}_i = 5$ and $E_k(1 - E_k/n_k) = 5(1 - \frac{1}{2}) = \frac{5}{2}$ for every group of ten, so each term of $\hat{C}$ is $(O_k - 5)^2 / \frac{5}{2}$.
- A grouping in which every group contains five successes has $O_k = 5$ for all k and $\hat{C} = 0$, the smallest value a chi-squared statistic can take. A grouping in which five groups contain ten successes and five contain none has $(O_k - 5)^2 = 25$ in every group, so each term is 10 and $\hat{C} = 100$. Referred to $\chi^2(8)$, the first is an impossibly good fit and the second an impossibly bad one; no data set has changed between them.
- Every $\hat{\pi}_i$ is the same, so sorting by $\hat{\pi}_i$ leaves the order of the observations entirely to the tie-breaking rule, and any partition into ten blocks of ten is a possible result. The statistic is therefore not a function of the data and the fitted model: it also depends on the order in which the rows happened to be stored. Software makes that choice silently.
- No. The model is correct, and the whole range from 0 to 100 is available; the argument uses only that the fitted probabilities are tied, not that the fit is wrong. With a continuous covariate exact ties are rare, but near-ties reproduce the same instability in milder form, and that — together with the G -dependence of **Example (The same model, three verdicts)** and the absence of any proof behind the $\chi^2(G - 2)$ reference — is why $\hat{C}$ is a description and not a test.

Exercise. C2

Proposition 35.4.3 says the AUC ignores calibration. The converse question is what it measures when calibration holds. Let X be the covariate vector, $\pi(x) = \Pr(Y = 1 \mid X = x)$, $\mu = \Pr(Y = 1) \in (0, 1)$, and let the score be the *calibrated* one $\hat{s} = \pi(X)$. Write P_1 and P_2 for independent copies of $\pi(X)$.

- Show that the covariate of a uniformly chosen case has law $\pi(x) dF(x)/\mu$ and that of a

uniformly chosen control $\{1 - \pi(x)\} dF(x)/(1 - \mu)$, independently, where F is the law of X . Deduce that

$$\text{AUC} = \frac{E[P_1(1 - P_2)1\{P_1 > P_2\}] + \frac{1}{2} E[P_1(1 - P_2)1\{P_1 = P_2\}]}{\mu(1 - \mu)}.$$

- (b) Writing A , B and T for the three expectations $E[P_1(1 - P_2)1\{P_1 > P_2\}]$, $E[P_1(1 - P_2)1\{P_1 < P_2\}]$ and $E[P_1(1 - P_2)1\{P_1 = P_2\}]$, show that $A + B + T = \mu(1 - \mu)$ and $A - B = \frac{1}{2} E|P_1 - P_2|$, and hence that

$$\text{AUC} = \frac{1}{2} + \frac{E|P_1 - P_2|}{4\mu(1 - \mu)}.$$

- (c) Deduce that $\text{AUC} = \frac{1}{2}$ exactly when $\pi(X)$ is degenerate, and that $\text{AUC} = 1$ exactly when $\pi(X) \in \{0, 1\}$ with probability one.
- (d) Check the formula on the population in which $\pi(X)$ takes the values 0.4 and 0.6 with probability $\frac{1}{2}$ each, by computing the AUC both ways.

For a calibrated model the AUC is therefore a measure of how far apart the achievable probabilities are, and nothing else. What does that say about comparing two models by their AUCs?

Solution.

- (a) $\Pr(X \in dx \mid Y = 1) = \Pr(Y = 1 \mid X \in dx) dF(x) / \Pr(Y = 1) = \pi(x) dF(x) / \mu$, and likewise for a control; I and J are drawn independently, so the two covariates are independent. Substituting in Proposition 35.4.3 and writing the expectations against dF gives the display, the factor $\pi(x_1)\{1 - \pi(x_2)\} / \{\mu(1 - \mu)\}$ being the density of the tilted pair with respect to $dF \times dF$.

- (b) Independence gives $A + B + T = E[P_1(1 - P_2)] = E(P_1) E(1 - P_2) = \mu(1 - \mu)$. Exchanging the labels of P_1 and P_2 turns B into $E[P_2(1 - P_1)1\{P_1 > P_2\}]$, so

$$A - B = E[\{P_1(1 - P_2) - P_2(1 - P_1)\}1\{P_1 > P_2\}] = E[(P_1 - P_2)1\{P_1 > P_2\}] = \frac{1}{2} E|P_1 - P_2|,$$

the last step by symmetry. Hence $A = \frac{1}{2}\{\mu(1 - \mu) - T + \frac{1}{2} E|P_1 - P_2|\}$ and $A + \frac{1}{2}T = \frac{1}{2}\mu(1 - \mu) + \frac{1}{4} E|P_1 - P_2|$. Divide by $\mu(1 - \mu)$.

- (c) $E|P_1 - P_2| = 0$ iff $P_1 = P_2$ almost surely, that is iff $\pi(X)$ is almost surely constant, necessarily at μ . For the upper end, $A, B, T \geq 0$ and $A + B + T = \mu(1 - \mu)$ give $E|P_1 - P_2| = 2(A - B) \leq 2\mu(1 - \mu)$, with equality iff $B = T = 0$ and $A = \mu(1 - \mu)$. If $\Pr\{0 < \pi(X) < 1\} = q > 0$, then either the law of $\pi(X)$ has an atom at some $p \in (0, 1)$, in which case $T \geq p(1 - p) \Pr\{\pi(X) = p\}^2 > 0$, or it does not, in which case $\Pr\{P_1 < P_2, \text{ both in } (0, 1)\} = q^2/2 > 0$ and $B > 0$ because $P_1(1 - P_2) > 0$ there. Either way equality fails. Conversely if $\pi(X) \in \{0, 1\}$ almost surely then $E|P_1 - P_2| = 2\mu(1 - \mu)$.
- (d) Here $\mu = \frac{1}{2}$ and $|P_1 - P_2|$ is 0.2 with probability $\frac{1}{2}$ and 0 otherwise, so $E|P_1 - P_2| = 0.1$ and the formula gives $\frac{1}{2} + 0.1/1 = 0.6$. Directly: among cases $\Pr(P = 0.6) = 0.6$ and among controls $\Pr(P = 0.6) = 0.4$, so $\Pr(\hat{s}_I > \hat{s}_J) = 0.6 \times 0.6 = 0.36$ and the ties contribute $\frac{1}{2}(0.6 \times 0.4 + 0.4 \times 0.6) = 0.24$, for 0.60.

Two calibrated models on the same population are therefore ranked by the spread of their fitted probabilities, and an AUC that barely moves — 0.942 to 0.944 in Example (Voting intention in the 1996 American election) — says that the extra covariates barely spread them,

on a scale compressed by the factor $4\mu(1 - \mu) \leq 1$. That is one fact about a model, not a verdict on it: a covariate that moves a few individuals from 0.2 to 0.8 can matter enormously to those individuals and hardly at all to a rank statistic averaged over all pairs.